FRONT VIEW — TARGET FORM



~145×72×14 mm · ~220 g · NTS

mesh radio marks shown in blue

WHAT CHANGED VS MONOLITH

— Monolith = flagship research device (titanium spine, optical docks).

— Wrong phone for a rancher in a -35 °C truck cab or a fly-in community.

— ArcticPhone is rural-first: mesh-native, satellite SOS-ready, offline-first, repairable, community-owned.

— Every ArcticPhone IS a node — calls, texts, data ride the community mesh.

ARCHITECTURE OPTIONS (DECISION REQUIRED)

OPT	APPROACH	PROS	CONS
A	Rugged Android + ArcticChain mesh HAT	fast to market, app ecosystem	Android bloat, cert effort
B	Custom Linux + mesh SoC (integrated SX1262)	clean, sovereign, repairable	OS work, smaller ecosystem
C	Hybrid: Linux core + Android layer	best of both	longest dev time

CAPABILITY SET

CAPABILITY	IMPLEMENTATION
Mesh node	SX1262 LoRa 915 + signed envelopes — every phone extends the network
Satellite SOS	L-band messenger module now; own constellation later
Offline apps	maps, manuals, forms, messaging EN/FR — useful with zero bars
Cellular	LTE Cat-1/4 when towers exist — proven path from L1
Power	replaceable battery + USB-C PD + solar trickle option
Rugged	IP68 · -30..+55 °C · glove UI · 1.5 m drop target

ARCTICCHAIN — RURAL NODE FLEET

EMEBET INC. · emebetos.ca
SOLE PROPRIETARY OWNER — SURAFEL NEGUSSIE, PLANNER/MAKER

DRAWN BY **Surafel Negussie**
CHECKED BY
DATE
PROJECT

SCALE **NTS**
SHEET **1/2**
DWG NO **AF-P-101**
UNITS mm

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TARGET SPECIFICATION

PARAMETER	TARGET	PARAMETER	TARGET
Envelope	~145×72×14 mm	Mass	~220 g
Display	5–6in transflective	SoC	quad-core, LTE, GNSS
Mesh radio	SX1262 915 + MCU	Sat SOS	L-band module (rev-A)
Battery	replaceable ~4000 mAh	Charging	USB-C PD + solar
OS	Android Go OR Linux	UI	glove-friendly, EN/FR
Rating	IP68 · -30..+55 °C	Repair	standard screws, open back

BOM CATEGORIES (NOT FIXED — ARCHITECTURE PENDING)

CATEGORY	OPTIONS / NOTES
Compute+display	rugged Android module (A) OR custom board (B)
Mesh module	SX1262 + nRF52/ESP32 — same envelope as fleet
Sat messenger	inReach-class module, Iridium Certus OEM, or future own-sat
Cellular	LTE Cat-4 modem + SIM — proven path from L1 HAT
Power	4000 mAh LiFePO4 pouch, USB-C PD, solar pad option
Enclosure	PC+TPU bumpers, gasketed ports, captive screws

ROADMAP TO FIRST UNIT

— Phase 0 (now): architecture decision + industrial design — this sheet.
— Phase 1: prototype on dev-kit + mesh HAT — prove mesh+SOS on existing hardware.
— Phase 2: custom enclosure + integration — print → machine → mold decision.
— Phase 3: certification (ISED/RF) + pilot batch — 50–100 units for founding communities.
— Phase 4: volume — repairable design pays off at scale (modules, not glue).

UNIQUE VS EVERY OTHER PHONE

— The only phone that is a FIRST-CLASS MESH CITIZEN — your phone extends the network.
— Satellite SOS as a core feature, not an accessory — for places towers aren't.
— Offline-first: apps, maps, forms, messaging all useful with zero bars.
— Community-owned: the network your phone joins belongs to your community.
— Repairable by design — a local tech fixes it with hand tools.

RELATIONSHIP TO OTHER NODES

— Phone = personal node; L1 = field/shared; Totem = public; Mite = environmental; Anchor = infrastructure.
— All speak the same signed mesh — a call routes phone→phone→brick→satellite.
— Phone is the highest-volume node — its radio is the network's user terminal.

ARCTICCHAIN — RURAL NODE FLEET

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